

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1821

Leesburg Picciola 138 kV

April 21, 2023

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1821. The Interconnection Customer proposes a 500.0 MW battery energy storage facility interconnecting at the Leesburg Picciola 138 kV substation in the Virginia area. The request was received on July 20, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 485 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,767,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                     | Contingency                                            | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------|--------------------------------------------------------|-------------|--------------|----------|
| LEESBURG PICCIOLA – OROVADA 138kV Line                 | CLINCH RIV – CARLTON HILLS 345kV                       | 147.6       | 391          | 48.1     |
| LEESBURG PICCIOLA – HAYSI 138kV                        | CARLTON HILLS 345 kV/1 kV Autotransformer              | 144.4       | 276          | 48.9     |
| CARLTON HILLS – OROVADA 138kV L                        | ATOKA SUBSTATION – CARLTON HILLS 345kV                 | 137.6       | 463          | 37.2     |
| CARLTON HILLS 138/1kV Transformer                      | ATOKA – CARLTON HILLS 345kV                            | 136.4       | 481          | 37.2     |
| CARLTON HILLS 345/1kV Transformer                      | ATOKA – CARLTON HILLS 345kV line                       | 131.0       | 501          | 37.2     |
| TURKEY PEN – LEESBURG PICCIOLA SUBSTATION 138kV        | CARLTON HILLS – CLINCH RIV 345kV                       | 123.8       | 275          | 21.2     |
| LOCHMERE HYDROELECTRIC PLANT – HINTON AP 138kV         | MANATEE – HINTO AP 345kV line                          | 120.3       | 192          | 7.1      |
| MINNIX MOUNTAIN SUBSTATION – OPTICAL COATING LAB 138kV | KEAN MTN AND GRASSY CREEK – BARR LAKE SUBSTATION 345kV | 117.4       | 261          | 8.1      |
| RECWAY – MINNIX MTN SUBSTATION 138kV                   | KEAN MTN AND GRASSY CRK – BARR LAKE 345kV              | 117.2       | 265          | 8.1      |
| RECWAY – PRAIRIE CITY 138kV Line                       | KEAN MTN AND GRASSY CREEK – BARR LAKE SUBSTATION 345kV | 117.1       | 269          | 8.1      |
| LONEOME PINE SUBSTATION – PRAIRIE CITY 138kV           | KEAN MTN AND GRASSY CREEK – BARR LAKE 345kV L          | 116.5       | 288          | 8.1      |
| MUNICIPAL LIGHT – LONESOME PINE 138kV                  | KEAN MTN AND GRASSY CRK – BARR LAKE 345kV              | 116.0       | 310          | 8.1      |
| DAFFIN GIN – MUNICIPAL LIGHT SUBSTATION 138kV          | KEAN MTN AND GRASSY CRK – BARR LAKE 345kV Line         | 115.8       | 318          | 8.1      |

| Monitored Facility                                | Contingency                                            | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------|--------------------------------------------------------|-------------|--------------|----------|
| STA 11 – SPRINGBANK 138kV                         | KEAN MTN AND GRASSY CRK – BARR LAKE 345kV              | 115.4       | 483          | 9.2      |
| STOCK ISLAND – STATION 11 138kV                   | BARR LAKE – KEAN MTN AND GRASSY CRK 345kV              | 114.4       | 504          | 9.2      |
| HURLEY – DOOLEY 138kV                             | KEAN MTN AND GRASSY CRK – BARR LAKE 345kV              | 109.4       | 276          | 10.6     |
| LENA – OU SPIRIT WIND FARM 138kV L                | BARR LAKE – NEW HOPE AP 345kV line                     | 108.6       | 295          | 8.7      |
| LENA – OPTICAL COATING LAB 138kV                  | BARR LAKE – NEW HOPE AP 345kV                          | 108.5       | 298          | 8.7      |
| PEYTON – COVADONGA 1014-1013-1012-1011 138kV Line | LOWMOOR – HINTON AP 345kV                              | 107.9       | 326          | 5.0      |
| MOUNT VIEW – PEYTON 138kV Line                    | LOWMOOR – HINTON AP 345kV Line                         | 107.8       | 288          | 5.0      |
| OCONTO – SPRINGBANK 138kV L                       | LOWMOOR – HINTON AP 345kV                              | 106.9       | 305          | 5.7      |
| NO TROY – OCONTO 138kV                            | LOWMOOR – HINTON AP 345kV line                         | 106.8       | 290          | 5.7      |
| PITCHGEN – LOCHMERE HYDROELECTRIC PLANT 138kV     | MANATEE – HINTON AP 345kV L                            | 105.8       | 224          | 7.1      |
| BRIAR – PITCHGEN 138kV line                       | MANATEE – HINTON AP 345kV Line                         | 105.4       | 346          | 10.5     |
| BRIAR – FRASER 138kV                              | MANATEE – HINTON AP 345kV line                         | 105.3       | 349          | 10.5     |
| WINNEBAGO WIND POWER PROJECT – LEBANON 138kV      | BARR LAKE – KEAN MTN AND GRASSY CREEK SUBSTATION 345kV | 104.6       | 249          | 7.4      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                               | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| LEESBURG PICCIOLA – OROVADA 138kV Line                 | Rebuild line with higher-capacity conductor | \$3,093,000   | 36% (\$1,111,000)     |
| LEESBURG PICCIOLA – HAYSI 138kV                        | Rebuild line with higher-capacity conductor | \$19,179,000  | 38% (\$7,214,000)     |
| CARLTON HILLS – OROVADA 138kV L                        | Rebuild line with higher-capacity conductor | \$537,000     | 25% (\$136,000)       |
| CARLTON HILLS 138/1 kV TX                              | Replace transformer with higher-rated unit  | \$4,147,000   | 26% (\$1,056,000)     |
| CARLTON HILLS 345 kV/1 kV Autotransformer              | Replace transformer with higher-rated unit  | \$19,182,000  | 26% (\$4,884,000)     |
| TURKEY PEN – LEESBURG PICCIOLA 138kV line              | Rebuild line with higher-capacity conductor | \$22,682,000  | 48% (\$10,812,000)    |
| LOCHMERE HYDROELECTRIC PLANT – HINTON AP 138kV         | Rebuild line with higher-capacity conductor | \$3,104,000   | 11% (\$350,000)       |
| MINNIX MOUNTAIN SUBSTATION – OPTICAL COATING LAB 138kV | Rebuild line with higher-capacity conductor | \$3,066,000   | 11% (\$332,000)       |

| Facility                                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| RECWAY – MINNIX MTN SUBSTATION 138kV              | Rebuild line with higher-capacity conductor | \$4,781,000   | 11% (\$518,000)       |
| PRAIRIE CITY – RECWAY 138kV line                  | Rebuild line with higher-capacity conductor | \$1,135,000   | 11% (\$123,000)       |
| PRAIRIE CITY – LONESOME PINE 138kV L              | Rebuild line with higher-capacity conductor | \$3,725,000   | 11% (\$404,000)       |
| MUNICIPAL LIGHT – LONESOME PINE 138kV             | Rebuild line with higher-capacity conductor | \$7,067,000   | 11% (\$766,000)       |
| MUNICIPAL LIGHT – DAFFIN GIN 138kV                | Rebuild line with higher-capacity conductor | \$7,548,000   | 11% (\$818,000)       |
| SPRINGBANK – STA 11 138kV                         | Rebuild line with higher-capacity conductor | \$11,542,000  | 7% (\$796,000)        |
| STATION 11 – STOCK ISLAND 138kV L                 | Rebuild line with higher-capacity conductor | \$2,735,000   | 7% (\$189,000)        |
| HURLEY – DOOLEY 138kV                             | Reconductor line                            | \$156,000     | 23% (\$36,000)        |
| LENA – OU SPIRIT WIND FARM 138kV L                | Reconductor line                            | \$4,594,000   | 10% (\$446,000)       |
| LENA – OPTICAL COATING LAB 138kV L                | Reconductor line                            | \$4,317,000   | 10% (\$419,000)       |
| PEYTON – COVADONGA 1014-1013-1012-1011 138kV Line | Reconductor line                            | \$2,670,000   | 13% (\$354,000)       |
| MOUNT VIEW – PEYTON 138kV Line                    | Reconductor line                            | \$1,366,000   | 13% (\$181,000)       |
| OCONTO – SPRINGBANK 138kV                         | Reconductor line                            | \$1,798,000   | 7% (\$130,000)        |
| NO TROY – OCONTO SUBSTATION 138kV line            | Reconductor line                            | \$774,000     | 7% (\$56,000)         |
| LOCHMERE HYDROELECTRIC PLANT – PITCHGEN 138kV L   | Reconductor line                            | \$2,042,000   | 11% (\$230,000)       |
| BRIAR – PITCHGEN 138kV line                       | Reconductor line                            | \$1,123,000   | 11% (\$123,000)       |
| BRIAR – FRASER 138kV                              | Reconductor line                            | \$1,206,000   | 11% (\$132,000)       |
| LEBANON – WINNEBAGO WIND POWER PROJECT 138kV      | Reconductor line                            | \$739,000     | 20% (\$151,000)       |
|                                                   | Total allocated to project                  |               | \$31,767,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.